Software Requirements Specification

for

HomeZapp

Version 1.0 approved

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Group 8

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1. Introduction

HomeZapp is a platform that allows users to either buy or sell properties. These houses are categorized into the 6 major architectural aesthetics- Colonial, Victorian, Farm, Mediterranean, Craftsmen, Modern, and other. The users are divided into 3 major classes- buyers, administration, and sellers. The website will use PostgreSQL to store all the houses' information into a database, which would be accessible to the users through the website.

1.1 Purpose

The software requirements specification document is intended to provide a detailed report of the features, design and functionality of the product. The document also serves the purpose of communicating the software development team's planning to the board members and investors of HomeZapp. Additionally, this document can also be used as a point of reference for future employees and developers in case they want to make any modifications to the ecommerce platform.

1.2 Document Conventions

This document uses Times New Roman font throughout the entire document. Major header lines use bold, size 24 font. Subsections use bold, italics, size 18 font. Smaller subsections use bold, italic, size 13 font. Any part of this document that is not a header uses size 11 font.

1.3 Intended Audience and Reading

This document is intended for developers, project managers, marketing staff, users, testers, and documentation writers.

1.4 Product Scope

This is a website aiming to provide categorization and useful information to potential house buyers. It will benefit buyers because they can see lots of information about the homes and private sellers because the website will require them to give necessary information to make sure they are marketing themselves well. Our goal is to provide a complete process for home buyers and sellers trying to find/sell their home. Our objectives are to provide guidance to inexperienced home sellers and options for buyers. Our business goal is to make as many transactions as possible between buyers and sellers by matching customers with their desired house and providing important information about the house and seller, as well as information to the seller about potential buyers. Also by allowing for communication between the buyers and sellers.

1.5 References and Acknowledgements

- Zillow for website layout and functional reference- Zillow: Real Estate, Apartments, Mortgages & Home Values
- Django- The web framework for perfectionists with deadlines | Django
- Python- Welcome to Python.org
- PostgreSQL- PostgreSQL: The world's most advanced open source database

2. Overall Description

2.1 Product Perspective

HomeZapp is a new, self-contained digital project with functions designed to connect the buyer directly to the seller under the management of admins. There is no initial framework or system to build upon; this software must be built from scratch in two parts, with a frontend and backend. The frontend will allow the user to interact with the webpage using a built-in interface. In the backend, our software will allow us to store and aggregate data. The web application itself will be built on top of Django+Python.

As stated by the customer, this product must have users with different roles, the buyer should be able to search, compare, cancel the process for, and buy different types of products, a seller should be able to add, sell, and receive payments for their products; and an admin should be able to approve/black new user accounts and products.

Sellers must be verified to be able to post listings. The verification process takes place using another system. This system must be compatible with the verification system. The transactions and majority of the communication between buyers and sellers happens off site.

2.2 Product Functions

The product -

- Connects buyers to sellers and vice versa.
- Make sure that the buyer has all information needed to make a decision about the house.
- Allow reports to be made by users about listings
- Makes it so the buyer or the seller can pull out of the house application/purchasing process
- Handles the actual payment from buyer to seller
- Lets users browse houses based on architectural features and not just prices
- Provides educational information about the different architectural types to help users make an informed decision

2.3 User Classes and Characteristics

The main expected user classes for this project are home buyers, home sellers, and administration. Home buyers will be identifiable internally through their email, which can be used to block malicious accounts by administration. Home sellers can post, edit, and delete their listings. Once a buyer has found a listing, they can communicate with the seller and pay the agreed upon price. Administrative

users can remove listings and block buyer and seller accounts. The buyer and seller users are the most important.

2.4 Operating Environment

Backend: We chose Django, since it is a Python framework, a language that we are already familiar with and have great experience in. This makes our application easier to maintain and further add stuff if and when needed.

Database: We chose Postgresql for the database because of the reliance on relational properties in our product. Since Postgresql is best at handling complex relations, it makes it well suited for our website.

Frontend- We chose HTML/CSS since we are already familiar with those and also to keep our frontend simple and manageable.

It must also be compatible with verification software.

2.5 Design and Implementation Constraints

Since we are mostly accommodating to American users, the website is only available in English, so we will not be able to accommodate people who are more comfortable in other languages. The verification software for sellers is not included in the website program.

3. System Features

3.1 Create Account

3.1.1 Description and Priority

Priority: High

New users who navigate to the website can create an account to be stored within our database. This allows them to access buyer features or be verified to access seller features.

3.1.2 Stimulus/Response Sequences

Sequence 1

1) User navigates to the website
2) User selects “Create Account” on the homepage
3) System redirects user to create account page
4) User inputs required information, selects buyer account type, & submits form
5) System creates account and stores within its database
6) System logs the user in and redirects them to the buyer user dashboard.

Sequence 2

1) User navigates to the website
2) User selects “Create Account” on the homepage
3) System redirects user to create account page
4) User inputs required information, selects seller account type, & submits form
5) System redirects to external seller verification process in a new tab
6) External seller verification process alerts system verification was successful
7) System creates seller account and stores within its database
8) System redirects the user from the create account page to the seller user dashboard.

Sequence 3

1) User navigates to the website
2) User selects “Create Account” on the homepage
3) System redirects user to create account page
4) User inputs required information, selects seller account type, & submits form
5) System redirects to external seller verification process in a new tab
6) External seller verification process alerts system verification was not successful
7) The user is notified on the create account page the verification process was not successful.

Sequence 4

1) User navigates to the website
2) User selects “Create Account” on the homepage
3) User inputs required information incorrectly, selects either seller or buyer account type, & submits form
4) System cannot verify information
5) The user is notified of incorrect credentials

3.1.3 Functional Requirements

REQ-1: The system should let users register using the required information.

REQ-2: The system should prevent duplicate account creation.

REQ-3: The system should communicate with the external verification software

3.2 Login

3.2.1 Description and Priority

Priority: Medium

The users will be prompted to login to their account using their email and password. On a successful login attempt, the user will be signed into their account and redirected to their buyer/seller homepage.

3.2.2 Stimulus/Response Sequences

Sequence 1

1) The user selects the login option from the HomeZapp homepage
2) The system redirects the user to the login page
3) The user enters their email and password associated with their account
4) The system verifies the email and password as correct, signs the user in to their account, sees that they are a buyer, and redirects them to the homepage

Sequence 2

1) The user inputs incorrect email/password
2) The system notifies the user of this and does not sign them in

Sequence 3

1) The user inputs correct email/password
2) The system verifies the login as correct, logs them into their accounts, sees the user is a seller, and redirects them to their seller dashboard

3.2.3 Functional Requirements

REQ-1: The system allows users to log in using an email address and password from the stored database

REQ-2: The system displays an error message if the user enters incorrect information

REQ-3: The system sends users to the correct dashboard type tied to their user class(buyer/seller)

3.3 Seller Verification

3.3.1 Description and Priority

Priority: Medium

Users intending to sell a house must complete a verification process before posting listings. This allows the platform to be more secure

3.3.2 Stimulus/Response Sequences

Sequence 1

1) User selects “Seller verification” on house listing creation page
2) The system redirects user to seller verification page
3) The user inputs required information and submits the request
4) System forwards information to our external verification source
5) The system forwards approval back to the user.

Sequence 2

1) User inputs bad information
2) System informs the user they were not properly verified

3.3.3 Functional Requirements

REQ-1: The system should allow seller verification information to be submitted

REQ-2: The system should not allow users who are not sellers to access listing features until verification is approved.

3.4 Update Account information

3.4.1 Description and Priority

Priority: Low

Users can edit their account information after they have logged in.

3.4.2 Stimulus/Response Sequences

Sequence 1

1) User goes to account settings
2) The system redirects the user to account settings page
3) User selects “Edit profile” and submits information
4) The system validates the information and updates the data.

Sequence 2

1) User inputs bad information
2) System validates information as bad and informs user

3.4.3 Functional Requirements

REQ-1: The system should let users edit and update stored account information

REQ-2: The system should validate and save the data

3.5 Search for listings

3.5.1 Description and Priority

Priority: High

Users can search for listings based on their requirements and needs. This is a high priority feature and sits at an 8 out of 10, since without this feature the users would just be looking at random listings and wouldn’t be able to narrow down to their requirements.

3.5.2 Stimulus/Response Sequences

1) User goes to homepage and clicks on search bar
2) User types in what kind of house they want in their city of preference and/or bed/bath requirements
3) The system filters the houses from the database and shows the user their preferences
4) User can now view all of the options available

3.5.3 Functional Requirements

REQ-1: The system should allow users to search for criteria (Location, architectural style, price, etc.)

REQ-2: The system should be able to retrieve and display listings that match the criteria given by the user.

3.6 Filter listings

3.6.1 Description and Priority

Users can sort and filter listings based on what they are looking for and filter based on prices, location proximity, and relevance. It is a low-priority task and is not essential for the website to work, but it would improve the user experience. Because of this, it is a medium priority.

3.6.2 Stimulus/Response Sequences

1) The user searches for an item on the search bar
2) The system gives the user the available houses that match the search
3) The user chooses the sort option that they want from the dragdown options (Price:low to high, Price:high to low, Newest)
4) The system sorts according to the user's preferences
5) The user applies filters like location, price, beds and baths, and architectural types
6) The user clicks on the search button after choosing all the filter and sort options
7) The system updates the website to show the intended listings to the user

3.6.3 Functional Requirements

REQ-1: The system should allow users to apply sorting and filtering options

REQ-2: The system should update and display the listings based on the user selected filters

3.7 Remove Listing

3.7.1 Description and Priority

Priority: High

An admin can remove a listing that is no longer available. Listings can be taken down if they are suspected or confirmed to be fraudulent. This ensures that users see legitimate listings and maintain the integrity and trustworthiness of the platform.

3.7.2 Stimulus/Response Sequences

Sequence 1

1) Admin selects listing to remove
2) Admin confirms removal
3) System removes listing
4) System records the action
5) System notifies owner

Sequence 2

1) Admin tries to remove listing already removed
2) System informs admin of Error: listing not found or already removed

3.7.3 Functional Requirements

REQ-1: The system should allow admin users to remove listings from the website.

REQ-2: The system should log the removal and notify the listing owner.

3.8 Buyer Application Form Submission

3.8.1 Description and Priority

Priority: High
When the buyer is interested in a property, they can show interest to the seller by filling out the buyer application form and submitting it. This form includes legal documents and paperwork required for the seller's viewing.

3.8.2 Stimulus/Response Sequences

Sequence 1

1. The buyer views a listing and decides that they are interested in it. The buyer fills out the Buyer application form and uploads their legal documents and paperwork in there including- government issued ID, mortgage pre-approval documents, and bank statements. They then submit the form
2. The system saves the form and displays it in the listing's seller's dashboard

Sequence 2

1. The buyer enters their information and documents
2. The system recognizes documents are missing/bad information was given and alerts the buyer

3.8.3 Functional Requirements

REQ-1: The system should allow buyers to enter required details into the buyer application form

REQ-2: The system should validate file uploads in PDF, JPG and PNG formats

REQ-3: The system should be able to recognize bad information and inform the user

3.9 Accept/ Reject/ Negotiate Deal

3.9.1 Description and Priority

Priority: High

3.9.2 Stimulus/Response Sequences

Sequence 1- *(Accept)*

1. A buyer receives a notification from the seller saying their buyer application has been accepted and the seller sends them the legal contract to sign which includes the final price of the listing
2. The buyer fills over the contract and signs it with their legal name and the contract gets sent to the seller
3. The seller looks over at the legal contract and decides to move forward with it, and clicks on the finalize deal page
4. The buyer receives a notification on their notifications tab that when clicked on, takes them to the payment page. The buyer then fills out their payment details and the transaction is completed

Sequence 2- (Reject)

1. The buyer submits the buyer application form
2. The buyer receives a notification saying their application has been rejected for a reason that'd be stated by the seller
3. The buyer gets to apply one more time if it's an issue with their documents, but that would be their last chance to apply for the same listing

Sequence 3- (Negotiate)

1. The buyer submit the buyer application form for the listing that they are interested in
2. The buyer receives a notification saying their application has been accepted for that listing and the seller sends them a legal contract to sign
3. The buyer sees the notification on their notifications tab, but instead of filling out the contract they click on the negotiate button and type in their desired price value
4. The negotiation goes forward to the seller- and they make a decision whether to accept or reject deal

3.9.3 Functional Requirements

REQ-1: The buyer should be able to negotiate with the seller in case they would want to

REQ-2: Both the buyer and seller can backout (return) at any point in the deal before the seller clicks on finalize deal button

3.10 Payment Page

3.10.1 Description and Priority

Priority: High

After negotiations, buyers will be navigated to a payment page to submit payment. At this step, both parties are unable to back out of the selling process

3.10.2 Stimulus/Response Sequences

Sequence 1

1) After negotiations, the buyer chooses to navigate to "Pay and Sign."
2) The system redirects user to Payment Page
3) User completes the process by uploading their payment information
4) The system processes the transaction from the buyer to the seller and sends notifications to both that the payment has been submitted and the process is finished

Sequence 2

1) The buyer inputs bad information
2) The system informs the buyer of the bad information

3.10.3 Functional Requirements

REQ-1: The system should be able to handle transactions

REQ-2: The system should be able to send the notification

REQ-3: The system should be able to recognize bad payment information and inform the buyer

3.11 Ban User

3.11.1 Description and Priority

Priority: Medium

Admin is able to ban a user if the user is found to be a fraudulent and/ or if they break the terms of agreement

3.11.2 Stimulus/Response Sequences

1) A user has submitted a complaint about a listing
2) The admin views the complaint and decides to ban the user based on the context
3) The system records that the user is banned and will not allow the user back into their account. Attempts to make a new account with the same email are also prevented by the system.

3.11.3 Functional Requirements

REQ-1: The system should allow admin users to ban user accounts that violate our rules

REQ-2: The system should disable the banned user's access to the platform
REQ-3: The system should be able to prevent attempts to create new accounts with the same email

3.12 Send notifications to buyers or sellers

3.12.1 Description and Priority

Admin is able to send alerts or notifications to the buyers and sellers through the website. It is a medium-priority task since it's not required for the website to work but it's also not a frivolous feature.

Priority: Medium

3.12.2 Stimulus/Response Sequences

1.) The website has a notifications tab that takes the users to a notifications inbox
2.) The notifications can vary from- new features to the website, update in terms of agreement and alerting a user of their interaction with a fraud user
3.) Whenever the users login, they can view their personalized notifications on the notifications inbox

3.12.3 Functional Requirements

REQ-1: The system should allow admins to create and send notifications

REQ-2: The system should store notifications and display them in the user's notification box.

3.13 Send Complaints

3.13.1 Description and Priority

Priority: High

This feature allows users to submit complaints about suspicious behavior. Although sellers will be verified before being able to sell, there could still be a chance of a bad actor. This gives users the

ability to help us catch those bad actors, while also giving sellers the ability to tell us about buyers that seem shady. Complaints will be sent to administrative users.

3.13.2 Stimulus/Response Sequences

1) A signed in user selects the “Report” option
2) The user submits their complaint form
3) The system then records the complaint and sends it to every admin user’s notification box

3.13.3 Functional Requirements

REQ-1: The system needs to be able to record complaints

REQ-2: The system should allow users to submit complaints regarding any user or listing

3.14 Wishlist listings

3.14.1 Description and Priority

Priority: High
This feature allows users to wishlist property listings for future reference. Saved listings will help users easily revisit a home they were interested in without needing to search again, which improves the general flow and feel of the website.

3.14.2 Stimulus/Response Sequences

Sequence One:

1) The user will sign in to their account
2) The system validates information and signs the user in, then redirects to the homepage
3) The user will browse the listings, and then select the save listing option on something they’re interested in
4) The system records the listing in that buyer’s saved listings collection
5) The user can then access their saved listings later.

Sequence Two:

1) User navigates to website
2) System redirects them to homepage
3) The user will browse the listings, and then select the save listing option on something they’re interested in
4) The system informs the user of Error: You must be signed in to save listing.

3.14.3 Functional Requirements

REQ-1: The system should allow a signed-in user to save property listings

REQ-2: The system should be able to store saved listings in the user's account.

3.15 Create listings

3.15.1 Description and Priority

Priority: High

This feature will allow sellers to create a property listing on the website. Sellers must provide required details about the property, such as location, description, price, images, and architectural style. The seller's ability to create listings is important for connecting buyers and giving buyers something to browse on the website.

3.15.2 Stimulus/Response Sequences

Sequence One:

1) The seller logs into their seller-verified account
2) The system validates the information and redirects user to homepage
3) The seller navigates to the "Create Listing" tab
4) The system redirects user to "Create Listing" tab
5) The seller enters property details (address, price, description, images, architectural style) and submits the information
6) The system stores the listing in the database and publishes it.

Sequence Two:

1) The seller logs into their seller-verified account
2) The system validates the information and redirects seller to homepage
3) The seller enters the property details of an already existing listing
4) The system informs the seller of Error: Listing already exists

3.15.3 Functional Requirements

REQ-1: The system allows sellers to create property listings

REQ-2: The system should require sellers to provide listing details before being able to publish anything

3.16 Manage listings

3.16.1 Description and Priority

Priority: High

This feature allows sellers to manage their listings on the website. Sellers will be able to edit, add, or remove listings as they see fit, ensuring accurate and up-to-date information.

3.16.2 Stimulus/Response Sequences

Sequence One:

1) The seller logs into their verified account
2) The system verifies information and redirects them to homepage
3) The seller navigates to their listings page
4) The system redirects the seller to listings page
5) The seller selects edit listing
6) The system redirects seller to edit listing page
7) The seller updates the listing information and submits the change
8) The system validates the saves and updates the listing

Sequence Two:

1) The seller logs into their verified account
2) The system verifies information and redirects them to homepage
3) The seller navigates to their listings page
4) The system redirects the seller to listings page
5) The seller updates the listing information with information already present on the listing and submits change
6) The system informs the seller of Error: The information entered is already present in database

3.16.3 Functional Requirements

REQ-1: The system should allow users that are signed in to edit the information of listings they have created

REQ-2: The system should allow sellers to remove their own listings.

3.17 Notification to Seller

3.17.1 Description and Priority

Priority: Medium

This feature enables interactions between users without requiring monetary transactions on the website. A buyer interested in a listing can start the

3.17.2 Stimulus/Response Sequences

Sequence One:

1) A buyer sees a property listing they're interested in
2) The buyer selects the "Notify seller" option
3) The system navigates buyer to "Notify seller" page
4) The buyer submits a message request, giving their email or phone number
5) The system records the request and sends a notification to the seller of the listing through the platform
6) The seller receives the notification and reaches out

Sequence Two:

1) A user who is not signed in sees a property listing they're interested in
2) The user selects notify seller option
3) The system informs the user of Error: You must be signed in to notify seller.

3.17.3 Functional Requirements

REQ-1: The system should allow users that are signed in to send a notification or message to a seller through a listing page

REQ-2: The system should be able to deliver a notification to the seller and store the record in the database

3.18 Log Out

3.17.1 Description and Priority

Priority: High

This feature allows all users to log out of their account, keeping it secure.

3.18.2 Stimulus/Response Sequences

1) A user of any kind(buyer/seller/admin) clicks log out
2) The system logs them out so their computer no longer has access until they log in again

3.18.3 Functional Requirements

REQ-1: The system can make it so users have to sign in again before they can access their account

REQ-2: The system should provide users with a logout option from their account page

3.19 Admin Login

3.19.1 Description and Priority

Priority: High

This feature enables admins to log into their account

3.19.2 Stimulus/Response Sequences

Sequence One:

1) An admin user on the admin login page inputs their information correctly
2) System logs them into their account and redirects them to the admin dashboard

Sequence Two:

1) An admin user on the admin login page inputs their information incorrectly
2) The system notifies the user of this

3.19.3 Functional Requirements

REQ-1: The system should log users in and store a cookie so that they are recognized until they log out and the cookie is deleted

REQ-2: The system shall notify users if their information is incorrect

3.20 Admin Create Account

3.20.1 Description and Priority

Priority: High

Admin can create an account to be stored within our database. This allows them to access admin options such as ban users, delete listings and send notifications to the user class.

3.20.2 Stimulus/Response Sequences

Sequence 1

1) When the admin class types website url/ admin they are taken to the admin login page
2) On the admin login page, the admin class can click on the ‘admin create account’ option
3) The system redirects the user to the admin create account page
4) After getting redirected to the admin create account page, the admin can enter their email ID and if it’s verified with the system, they will get a verification code on their email ID
5) If their email ID is a part of the system, they are able to create a password for their email ID, after which they will get redirected to the admin login page

Sequence 2

1) When the admin class types website url/ admin they are taken to the admin login page
2) On the admin login page, the admin class can click on the ‘admin create account’ option
3) The system redirects the user to the admin create account page

4) After getting redirected to the admin create account page, the admin can enter their email ID and if it's verified with the system, they will get a verification code on their email ID
5) If their email ID is not a part of the system, the website will say email is not verified.

3.20.3 Functional Requirements

REQ-1: The system shall have a list of emails that are the verified admins so that those admins can create an account and login
REQ-2: The system shall tell the admin if their email is not verified yet and thus they wouldn't be able to create account

4. Other Nonfunctional Requirements

4.1 Performance Requirements

1) Speed and Responsiveness- The home page, filter pages, and search pages must be loaded and interactive relatively fast
2) Capacity- the website is able to support 10,000 users simultaneously without quality degradation

4.2 Safety Requirements

1) Age verification- User must be at least 18 years old to be able to create an account and they would need to verify their age through their email
2) Hate speech removal- Any user that participates in hate speech can be flagged for hate speech and the system shall immediately send them a warning to stop using hate speech.

4.3 Security Requirements

1) Handling with Fradulents- the admin can remove fraud listing or users if they realize upon inspecting any reports made that the situation was a fraud
2) Password requirements- All passwords must be encrypted while registering as a logged in user to ensure the safety of both the users and the platform

4.4 Software Quality Attributes

1) Performance- When the user prompts a search, the results are returned within 5 seconds
2) Compatibility- The website will be able to run smoothly on most web browsers including chrome, safari, and firefox